

BENGBU, China

WMO: 582210

Lat: 32.8520N

Lon: 117.3207E

Elev: 28

StdP: 100.99

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.7	-3.3	-15.6	1.0	0.5	-12.9	1.2	0.9	7.6	3.5	7.0	3.9	1.8	340	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	35.6	26.6	34.4	26.2	33.1	25.5	28.4	33.0	27.8	32.2	27.2	31.4	3.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.1	31.2	26.6	22.3	30.8	26.1	21.5	30.2	92.3	33.3	89.5	32.3	86.7	31.6	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.2	5.5	-7.4	37.5	1.7	1.2	-8.6	38.3	-9.6	39.0	-10.5	39.7	-11.7	40.6		
			-8.8	29.1	1.7	0.9	-10.0	29.7	-11.0	30.2	-12.0	30.7	-13.2	31.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	2.7	5.2	10.2	16.5	21.8	25.7	28.5	27.5	23.3	18.1	11.1	4.8	
	DBStd	9.51	3.22	3.86	4.09	3.84	3.26	2.80	2.77	2.84	2.81	3.28	4.09	3.28	
	HDD10.0	613	227	140	47	2	0	0	0	0	0	34	163		
	HDD18.3	1874	485	367	252	79	8	0	0	1	44	219	418		
	CDD10.0	2929	1	7	54	197	366	471	573	399	250	67	3		
	CDD18.3	1148	0	0	1	24	115	221	314	285	151	36	2	0	
	CDH23.3	11365	0	1	14	198	936	2139	3753	3118	1019	179	9	0	
	CDH26.7	4624	0	0	1	47	316	872	1757	1316	286	30	1	0	
Wind	WSAvg	2.7	2.6	2.8	3.1	3.0	2.9	2.8	2.7	2.6	2.5	2.4	2.4	2.4	
Precipitation	PrecAvg	967	28	39	60	57	81	147	214	142	82	49	43	25	
	PrecMax	1475	95	87	207	146	218	391	473	286	184	223	134	78	
	PrecMin	615	1	7	6	7	4	34	86	35	1	0	0	0	
	PrecStd	214	25	23	43	34	53	94	109	63	54	52	36	21	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	19.4	25.2	30.9	34.2	36.3	37.3	37.0	34.1	29.4	24.6	16.7	
		MCWB	8.6	12.1	16.2	20.8	20.8	23.6	27.5	27.3	25.8	19.4	14.8	9.9	
	2%	DB	11.8	15.9	22.0	27.8	32.0	34.3	36.0	34.9	31.3	27.2	21.3	13.7	
		MCWB	6.8	9.7	14.6	19.1	20.3	23.3	27.4	27.5	24.2	18.7	14.1	8.7	
	5%	DB	9.6	13.4	19.6	25.6	30.1	32.7	34.7	33.5	29.6	25.3	19.2	11.7	
		MCWB	5.6	8.3	13.2	17.6	19.7	22.8	27.3	27.1	23.2	17.7	13.3	7.5	
	10%	DB	7.7	11.2	17.1	23.4	28.2	31.3	33.4	32.1	28.1	23.5	17.3	9.9	
		MCWB	4.6	7.2	11.2	16.5	19.4	22.7	26.9	26.6	22.1	17.1	12.4	6.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	14.0	18.9	22.7	24.0	27.0	29.4	29.1	27.5	22.6	17.7	11.6
			MCDB	12.9	17.6	22.4	27.5	28.5	31.3	34.2	33.8	32.3	26.3	20.3	13.9
2%		WB	7.8	11.0	15.7	20.7	22.7	26.1	28.6	28.3	26.1	20.7	16.0	9.8	
		MCDB	10.5	14.1	20.3	25.6	27.6	30.6	33.4	32.5	29.4	23.9	19.4	12.7	
5%		WB	6.3	9.2	13.8	18.9	21.8	25.3	28.1	27.7	24.9	19.6	14.5	8.4	
		MCDB	8.8	12.3	18.0	23.8	26.6	29.5	32.8	31.8	28.0	23.0	17.4	10.8	
10%		WB	5.0	7.8	12.2	17.4	20.9	24.5	27.5	27.2	23.7	18.5	13.2	7.2	
		MCDB	7.5	10.6	16.1	21.8	26.1	28.8	32.1	31.1	26.7	22.0	16.5	9.4	
Mean Daily Temperature Range		5% DB	MDBR	7.3	7.6	8.6	9.0	9.2	8.0	6.4	6.3	7.4	8.2	8.2	7.6
			MCDBR	11.0	11.6	12.5	12.4	12.4	10.6	8.2	8.0	9.0	10.9	11.1	10.7
	MCWBR		6.9	6.8	6.7	5.8	4.4	3.2	2.6	2.6	3.2	4.2	5.4	6.3	
	5% WB	MCDBR	9.5	9.5	11.3	10.5	9.9	7.7	7.6	7.0	7.3	8.3	8.5	8.7	
		MCWBR	6.4	6.2	7.0	6.1	5.0	3.1	3.0	2.8	3.4	4.1	4.9	5.7	
Clear-Sky Solar Irradiance	taub	0.596	0.657	0.672	0.670	0.682	0.776	0.738	0.731	0.691	0.697	0.642	0.599		
	taud	1.624	1.549	1.543	1.589	1.606	1.473	1.583	1.582	1.614	1.547	1.594	1.640		
	Ebn at Noon	578	595	636	667	666	604	625	619	616	557	535	544		
	Edh at Noon	210	249	269	267	264	301	269	266	247	246	215	197		
All-Sky Solar Radiation	RadAvg	2.25	2.62	3.81	4.56	4.86	4.63	4.65	4.37	3.78	3.18	2.45	2.15		
	RadStd	0.28	0.51	0.35	0.48	0.46	0.50	0.56	0.60	0.51	0.45	0.32	0.31		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (8 neighbors)	N/A	-0.75	N/A	N/A	N/A	N/A	+57	N/A	N/A	-49

Nomenclature: See separate page